LAB EXPERIMENT RECORD

Set 26

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	N LAHARI
Register Number	192473018
Date	20 October 2026
Day	Tuesday

Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

1. Histogram and Boxplot in R

Aim

To import the student dataset in R, plot a histogram for Math_Score and draw a boxplot of Science_Score by Gender with proper titles and labels.

R Code

R

```
data <- read.csv("Student_Mini_Data.csv")

data$Exam_Date <- as.Date(data$Exam_Date)

hist(data$Math_Score,

    main = "Distribution of Mathematics Scores",

    xlab = "Math Score",

    ylab = "Frequency",

    col = "skyblue",

    border = "black")

boxplot(Science_Score ~ Gender,

      data = data,

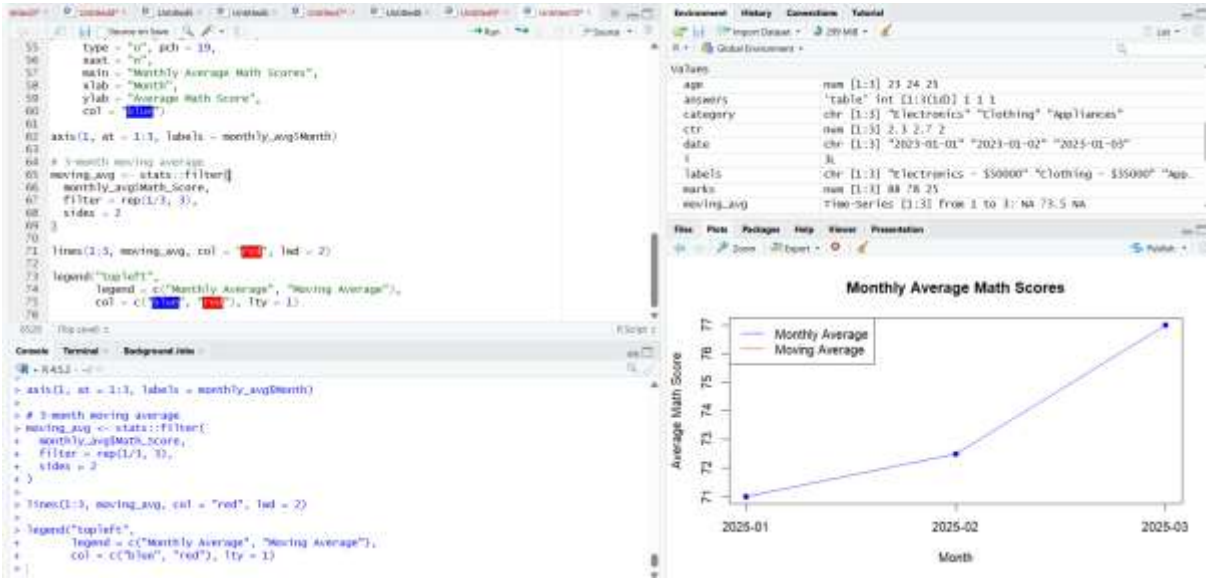
      main = "Science Scores by Gender",

      xlab = "Gender",

      ylab = "Science Score",

      col = c("orange", "lightgreen"))
```

Output



A histogram showing the distribution of mathematics scores and a boxplot comparing science scores by gender.

Observation

Mathematics scores range from 55 to 90. The boxplot helps compare the distribution and median science scores between male and female students. In this dataset, female students have higher science scores overall.

2. Scatter Plot: Study Hours vs Math Score

Aim

To create a scatter plot between Study_Hours and Math_Score using different colors for Gender and add a regression line.

R Code

R

```
data <- read.csv("Student_Mini_Data.csv")

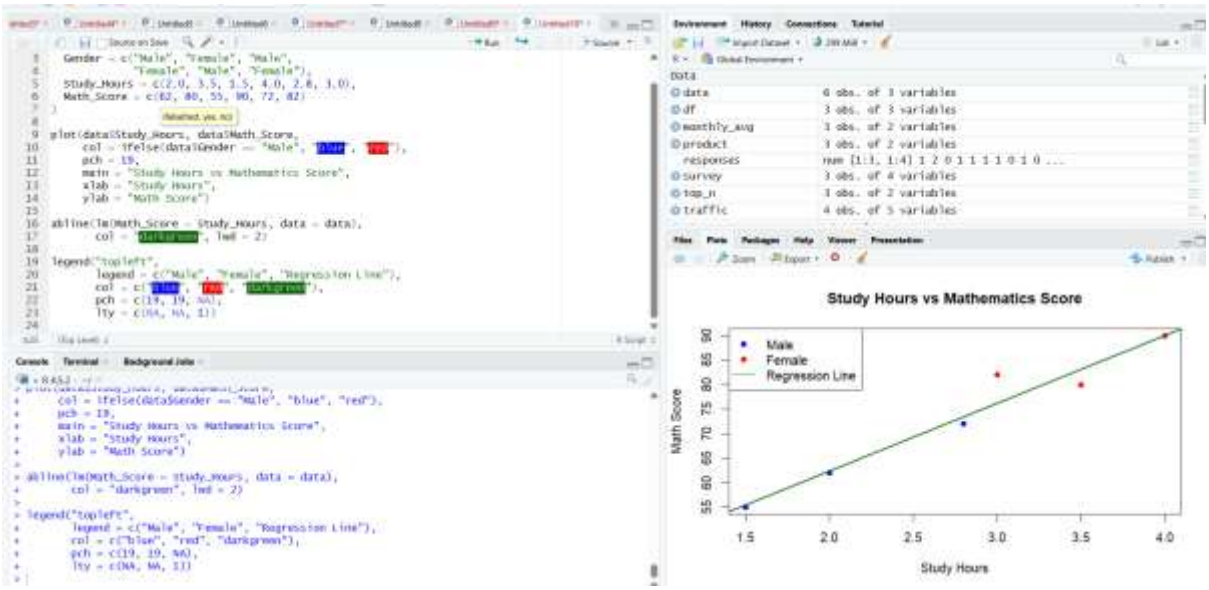
data$Color <- ifelse(data$Gender == "Male", "blue", "red")

plot(data$Study_Hours, data$Math_Score,

    col = data$Color,
```

```
    pch = 19,
    main = "Study Hours vs Mathematics Score",
    xlab = "Study Hours",
    ylab = "Math Score")
abline(lm(Math_Score ~ Study_Hours, data = data),
    col = "darkgreen", lwd = 2)
legend("topleft",
    legend = c("Male", "Female", "Regression Line"),
    col = c("blue", "red", "darkgreen"),
    pch = c(19, 19, NA),
    lty = c(NA, NA, 1))
```

Output



A scatter plot showing the relationship between study hours and mathematics scores, with separate colors for male and female students and a regression line.

Observation

The graph shows a positive relationship between study hours and mathematics scores. Students who study for more hours generally achieve higher scores in this sample. However, the dataset contains only six students, so the relationship should be interpreted cautiously.

3. Monthly Average Math Scores and Moving Average

Aim

To convert Exam_Date into Date format, calculate monthly average mathematics scores, plot a line chart, and apply moving average smoothing.

R Code

R

```
data <- read.csv("Student_Mini_Data.csv")

data$Exam_Date <- as.Date(data$Exam_Date)

data$Month <- format(data$Exam_Date, "%Y-%m")

monthly_avg <- aggregate(
  Math_Score ~ Month,
  data = data,
  FUN = mean
)

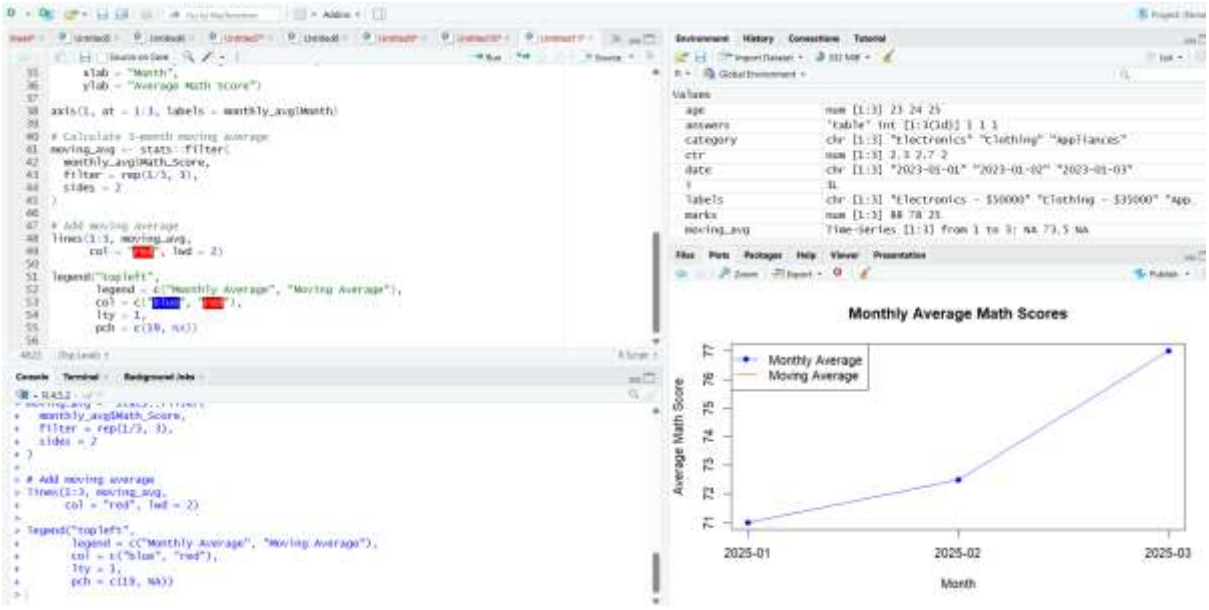
monthly_avg <- monthly_avg[order(monthly_avg$Month), ]

print(monthly_avg)

plot(1:nrow(monthly_avg), monthly_avg$Math_Score,
     type = "o",
     pch = 19,
     col = "blue",
     xaxt = "n",
```

```
    main = "Monthly Average Mathematics Scores",
    xlab = "Month",
    ylab = "Average Math Score")
axis(1, at = 1:nrow(monthly_avg),
    labels = monthly_avg$Month)
if (nrow(monthly_avg) >= 3) {
  moving_avg <- stats::filter(
    monthly_avg$Math_Score,
    filter = rep(1/3, 3),
    sides = 2
  )
  lines(1:nrow(monthly_avg), moving_avg,
        col = "red", lwd = 2)
  legend("topleft",
         legend = c("Monthly Average", "Moving Average"),
         col = c("blue", "red"),
         lty = 1,
         pch = c(19, NA))
}
```

Output



A line chart showing monthly average mathematics scores and a smoothed moving-average trend.

Observation

The monthly average mathematics scores are:

- January: 71
- February: 72.5
- March: 77

The scores show an overall increasing trend from January to March. The moving average smooths short-term variations and helps identify the general trend.

4. Tableau Dashboard: Student Performance Analysis

Aim

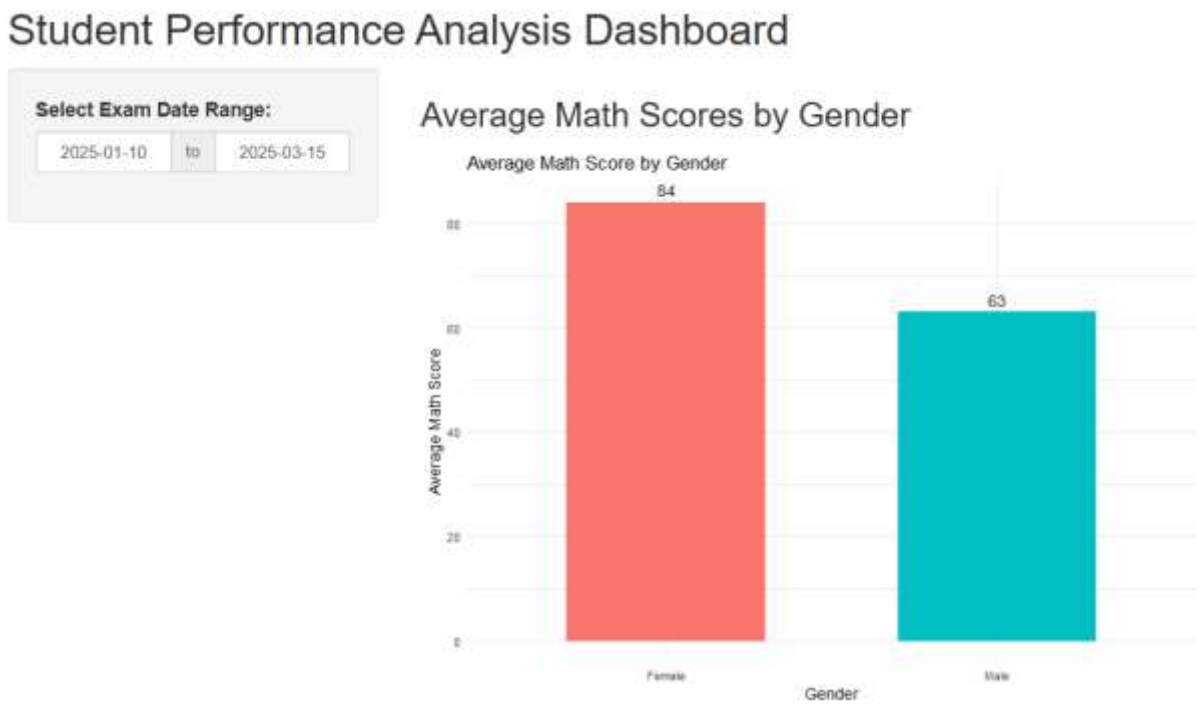
To create a Tableau dashboard displaying average mathematics scores by gender, score trends over time, and an interactive Exam Date filter.

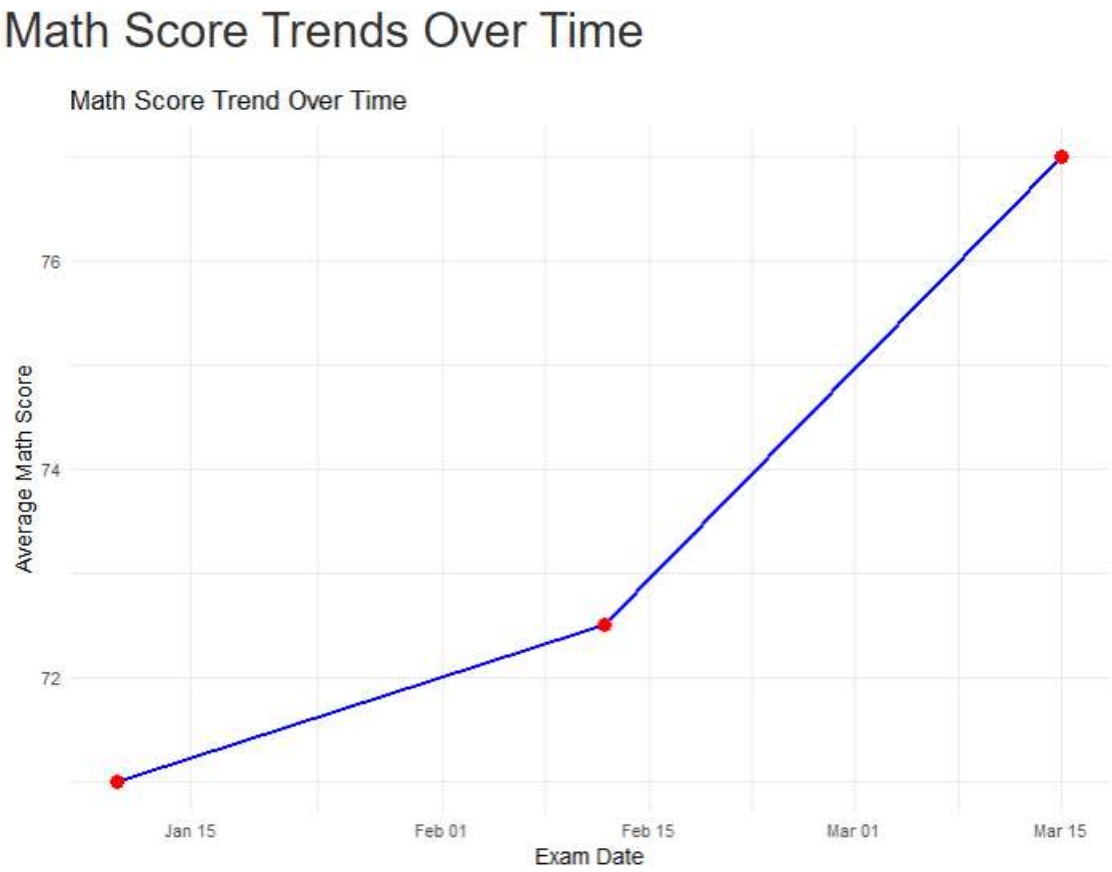
Tableau Procedure

1. Open Tableau and connect to Student_Mini_Data.csv.
2. Check that Exam Date is recognized as a Date field.

3. Create a new worksheet named **Average Math Score by Gender**.
4. Drag Gender to Columns and Math Score to Rows.
5. Change the aggregation of Math Score to Average.
6. Select Bar Chart and add suitable colours, labels, and a title.
7. Create another worksheet named **Score Trends Over Time**.
8. Drag Exam Date to Columns and Math Score to Rows.
9. Change Math Score to Average and select Line Chart.
10. Drag Exam Date to the Filters shelf and select **Show Filter**.
11. Select Dashboard → New Dashboard.
12. Add both worksheets to the dashboard and arrange them neatly.
13. Display the Exam Date filter and enable it to control both charts.
14. Add the title **Student Performance Analysis Dashboard** and save the dashboard.

Output





An interactive Tableau dashboard displaying average mathematics scores by gender and mathematics score trends over time, with an Exam_Date filter.

Observation

The dashboard helps compare mathematics performance between genders and explore changes in average scores across examination dates. The date filter allows users to view performance for selected dates.